



RV College of Engineering®
Department of Computer Science and Engineering
CIE - III

Course & Code	PARALLEL ARCHITECTURE AND GPU PROGRAMMING (CS372IA)		Semester: VII
Date: 15/12/2025	Duration: 120 minutes	Test Max. Marks: 50 Marks Quiz Max Marks: 10 Marks	Faculty In Charge: AN/SANS/MMN
USN :	Name :		

NOTE: Answer all the questions

Sl.no.	Questions	Marks	* BT	*CO
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QUIZ- 3

1	Name the three levels of parallelism in OpenACC..	2	2	4
2	Differentiate between Exploratory decomposition and hybrid decomposition	2	2	3
3	Give an example with detail steps for recursive decomposition	2	2	2
4	What is difference between Open ACC and CUDA?	2	1	1
5	Mention the characteristics of tasks in a parallel algorithm	2	2	1

TEST-3

1.	Explain the difference between task interaction graphs and task dependency graphs. Provide an example.	06	2	1
1b.	Compare static and dynamic mapping techniques for load balancing in parallel computing.	04	2	3
2.	Illustrate the concept of Open ACC data management using directives such as copy, copyin, copyout.	08	3	1
2.b	How does Open ACC handle reduction operations?	02	2	1
3.	Explain the key differences between the parallel and kernels constructs in OpenACC, providing examples of when to use each.	10	2	2
4.	Discuss different types of message passing operations in MPI, including blocking, non-blocking, buffered, and non-buffered communication. Provide suitable examples.	10	3	4
5.a	Illustrate any 4 parallel algorithm models in detail	10	3	3